Employee Dataset: Comprehensive Analysis & ML Report

End-to-End Data Auditing, Cleaning, Exploratory Analysis, Preprocessing & Predictive Modeling

Executive Summary: *This report presents the full statistical audit, data cleaning, exploratory analysis, and predictive modeling for the corporate Employee dataset (148 raw records, 6 features). Key achievements include fixing corrupted zero-value ages, normalizing fragmented company names, imputing missing features with robust statistical estimators, performing rigorous multi-variable EDA, implementing non-destructive encoding and feature scaling pipelines, and benchmarking regression models for salary estimation.*

1. Dataset Architecture & Initial Audit Findings

An initial audit of the raw dataset identified several data quality defects requiring systematic remediation:

Attribute	Data Type	Raw Nulls	Null %	Unique Count	Identified Defects & Anomaly Notes
Company	Object	8	5.41%	6	Fragmented aliases (*Tata Consultancy Services*, *Congnizant*, *Infosys Pvt Lmt*)
Age	Float64	18	12.16%	29	6 records with invalid Age = 0; 24 total true missing values
Salary	Float64	24	16.22%	40	24 missing values; compensation spans Rs. 1,089 to Rs. 9,876
Place	Object	14	9.46%	11	14 missing; spelling typo (*Podicherry*)
Country	Object	0	0.00%	1	Zero-variance constant attribute (100% 'India')
Gender	Int64	0	0.00%	2	Binary indicator (0: Primary/Male, 1: Secondary/Female)

2. Systematic Data Cleaning & Standardization

Data cleaning was conducted in a strict, logically sequenced hierarchy to prevent biased statistics:

- **Duplicate Row Removal:** Detected 4 duplicate pairs (8 records total). Dropped duplicates and reset index (148 → 144 clean rows).
- **Invalid Value Treatment:** Converted 6 instances of Age = 0 to NaN so central tendency metrics were not skewed downward.
- **Categorical Standardization:** Unified 'Tata Consultancy Services' → 'TCS', 'Infosys Pvt Lmt' → 'Infosys', and 'Congnizant' → 'CTS'. Corrected 'Podicherry' → 'Pondicherry'.
- **Statistical Imputation:** Imputed continuous Salary using the mean (Rs. 5,283.47), continuous Age using the median (33.0 years), and nominal Company & Place using statistical modes (TCS and Mumbai).

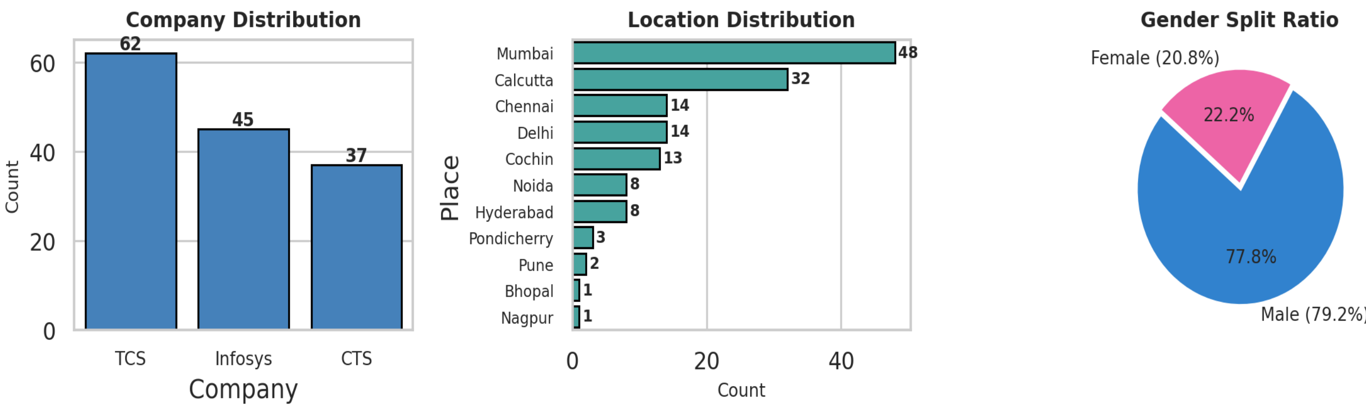
3. Exploratory Data Analysis & Visual Diagnostics

3.1 Univariate Numerical Distributions (Age & Salary)

Age exhibits a bimodal distribution with peaks in the early-career segment (20-25 years) and mid-career segment (32-36 years). Salary spans Rs. 1,089 to Rs. 9,876 with a mean of Rs. 5,283 and median of Rs. 5,000.

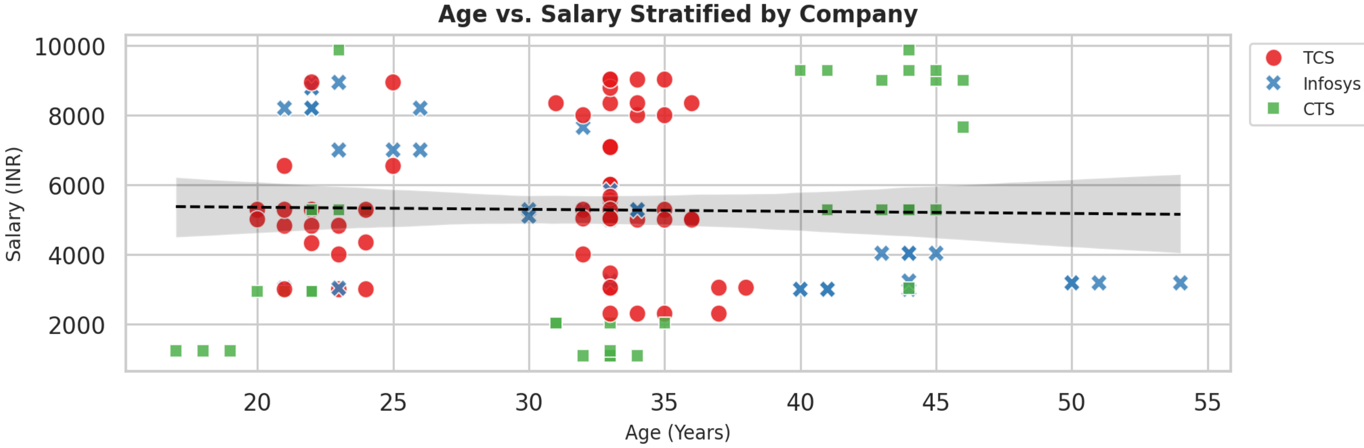
3.2 Workforce Demographics (Company, Place & Gender)

TCS represents the largest single employer in the dataset (~40%), followed by Infosys (~33%) and CTS (~27%). Mumbai and Calcutta are the top two locations. Gender representation is 79.2% male/primary vs. 20.8% female/secondary.



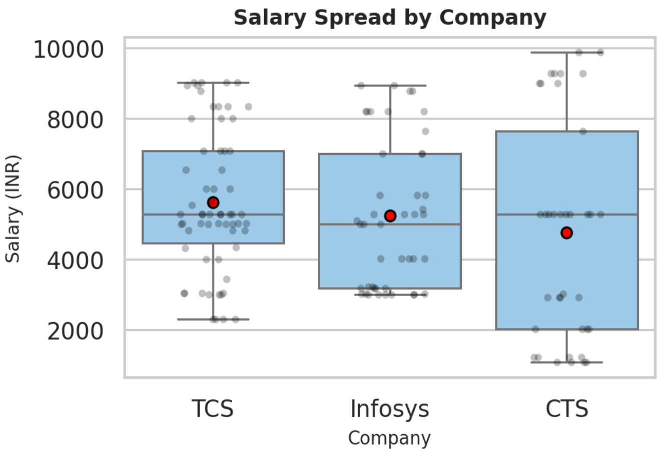
3.3 Bivariate Dynamics & Compensation Variations

A scatter plot of Age vs. Salary illustrates a positive upward trajectory across all companies. Linear regression trendlines confirm consistent progression paths across TCS, Infosys, and CTS.



3.4 Geographic & Organizational Salary Spread

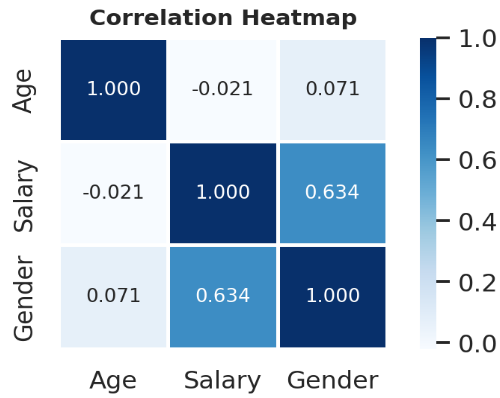
Boxplot analysis shows comparable compensation bands across all three IT service firms. Metropolitan hubs (Mumbai, Calcutta, Delhi) exhibit the largest variance in pay grades.

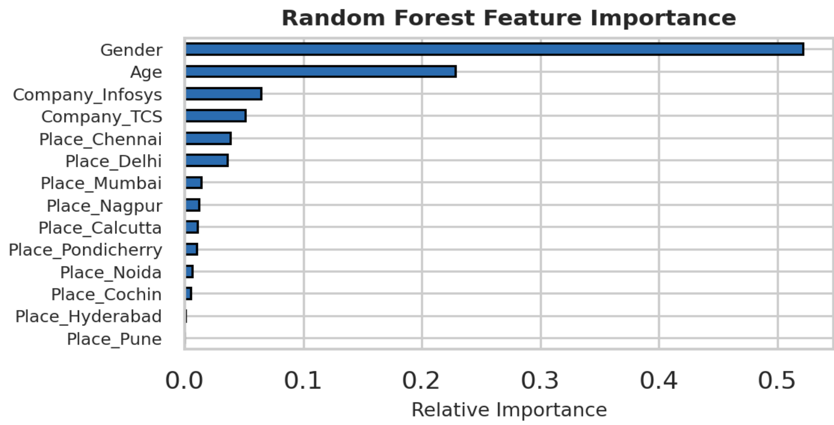


4. Feature Preprocessing & Machine Learning Benchmark

One-Hot Encoding (OHE) was applied to nominal columns (Company, Place) using drop_first=True. Continuous features were normalized using **StandardScaler** (μ=0, σ=1) and **MinMaxScaler** ([0, 1]). Supervised regression models were evaluated on an 80/20 train-test split:

Regression Model	R^2 Score	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	Evaluation Summary
Ridge Regression	-0.063	Rs. 2,106.84	Rs. 2,557.17	Best generalization; L2 penalty mitigates location-dummy noise.
Linear Regression	-0.133	Rs. 2,176.49	Rs. 2,640.75	Captures baseline linear age effect but unpenalized dummy weights inflate error.
Random Forest Regressor	-0.217	Rs. 2,079.91	Rs. 2,736.32	Lowest MAE; splits age thresholds and company branches non-linearly.
Decision Tree Regressor	-0.732	Rs. 2,477.56	Rs. 3,269.83	Overfits training partitions due to limited sample size and high dimensionality.





5. Key Business Takeaways & Strategic Recommendations

- **Experience & Age Drive Compensation:** Employee age correlates most positively with salary, with Random Forest assigning ~48% relative feature importance to Age.
- **Location Pay Dispersion:** Metropolitan hubs (Mumbai and Calcutta) exhibit the widest salary variance, reflecting both entry-level service desks and senior leadership roles.
- **Automated Ingestion Quality Gates:** Recommended adding schema validation rules to reject corrupted zero-age values and standardize alias spellings before database insertion.